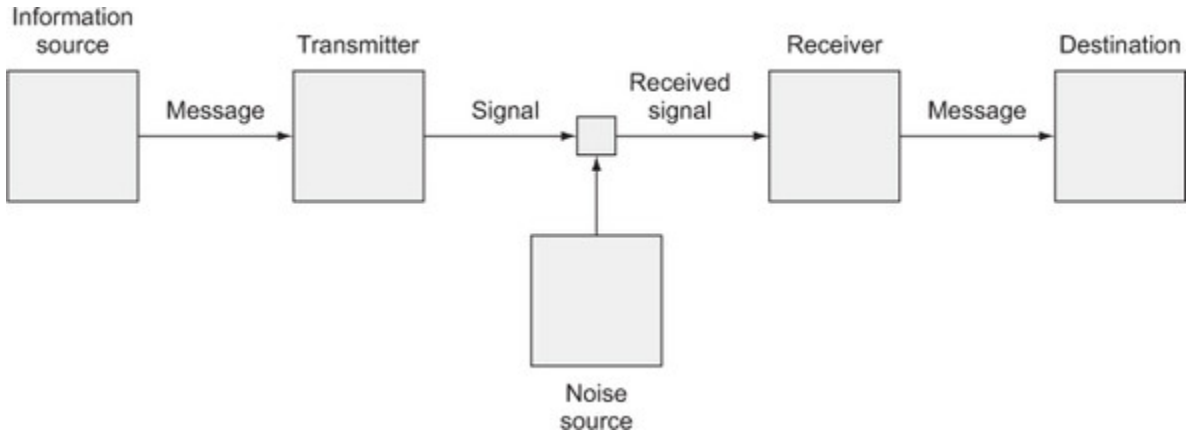
using radio waves), and we want to maximize the mutual information between *x* and the received message *y*. We do this by developing some way of encoding *x*, which might be a textual document, into a pattern of radio waves that minimizes the probability of the data being corrupted by noise. Once someone else receives the decoded message, *y*, they can be assured that their received message is very close to the original message.

Figure 8.22. Claude Shannon developed communication theory, which was born from the need to encode messages efficiently and robustly across noisy communication channels as depicted here. The goal is to encode the message such that the mutual information between the received message and the sent message is maximal.



In our example, *x* and *y* were both some sort of written message, but *x* and *y* need not be the same type of quantities. For example, we can ask what the mutual information is between the one-year stock price history of a company and its annual revenue: If we start with a very uncertain estimate about the annual revenue of a company, and then we learn the one-year stock price history, how much is our uncertainty reduced? If it's reduced a lot, the MI is high.

That example involved different quantities, but both used the units of dollars—that need not be the case either. We could ask what the MI is between the daily temperature and the sales of ice cream shops.

In the case of agent empowerment in reinforcement learning, the objective is to maximize the mutual information between an action (or sequence of actions) and the resulting future state (or states). Maximizing this objective means that if you know what action the agent took, you will have a high